there's nothing wrong with the strength of gravity. Maybe there's something wrong with the *universe* that's making gravity *seem* weaker than it really is.

What could make gravity seem weak? The solution might end up being surprisingly mundane. It's leaking. Into another dimension.

Here's how it works. As you're probably aware, we usually think of our universe as having three dimensions of space (east-west, north-south, up-down). In relativity, we count time as a dimension as well, and we talk about locations in a 4D spacetime (a position in space and some moment on the past-future continuum). In the *large extra dimensions* scenario, there is another direction, or several, that we can't access. All of the space part of our spacetime is limited to a 3D "brane"—think membrane—and a larger space extends outside it in some new direction (or directions) our limited human brains can only conceptualize mathematically. I should also mention that the "large" in "large extra dimensions" is a somewhat misleading term. Generally, if our universe does have extra dimensions, it might be effectively infinite in our usual three dimensions but extend no more than a millimeter in the new directions. (Imagine a large sheet of very thin paper—it's technically a three-dimensional object even though two of its dimensions are much larger than the third.) But to a particle physicist, accustomed to measuring distances that make atoms look big, millimeters may as well be miles. Accordingly, we refer to the extra space outside our own brane as the "bulk."

In this scenario, particle physics and gravity still act fundamentally differently from each other, but not because of their inherent strength. The difference is that all the forces of nature in particle physics—electromagnetism and the strong and weak nuclear forces—are confined to live on the brane. To them, the larger, higher-dimensional bulk doesn't exist. But gravity is not so limited. Gravity acts directly on spacetime, and that includes the spacetime outside of our 3D brane. So the gravity produced by a massive object in our own space loses a tiny bit of its apparent strength by leaking into the bulk, like an ink mark fading as it seeps into a sheet of paper. The fact that the new dimensions are so small compared to our ordinary ones means that this leakage isn't really noticeable until you're measuring the gravitational effects of things at millimeter distances, which is extremely hard to do. After all, most of the time, if you're near an object and you step a millimeter farther away from